Project Design Phase-I
Solution Architecture

Date	28 October 2026
Team ID	LTVIP2026TMID00420
Project Name	EV Market Analysis Using Tableau
Maximum Marks	4 Marks

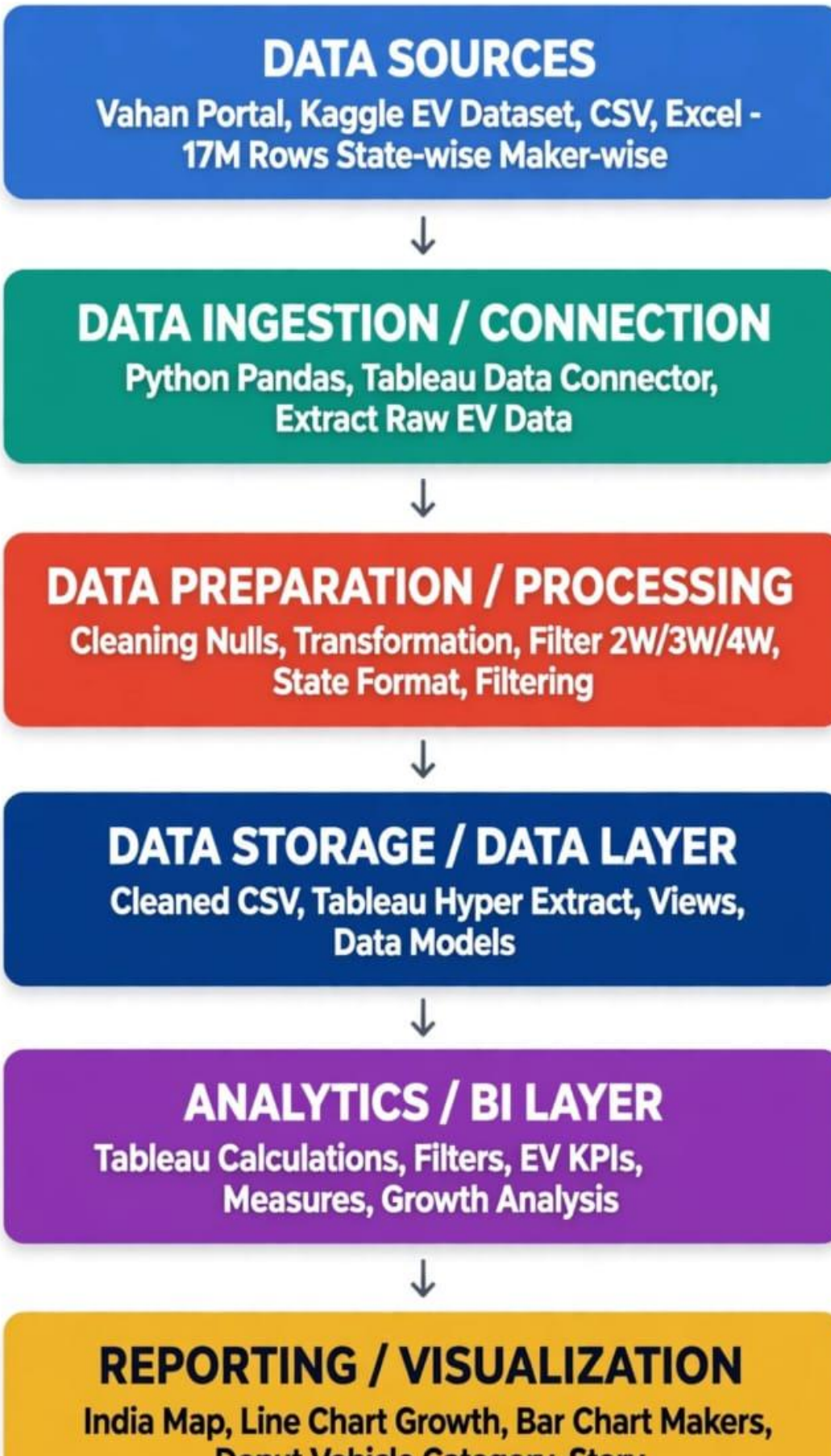
Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:

SOLUTION ARCHITECTURE DIAGRA



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Vahan Parivahan Portal & Kaggle se EV ka raw data lena, 17M rows state-wise, maker-wise, vehicle type	Vahan Portal, Kaggle, CSV, Excel
Data Ingestion / Connection	Raw EV data ko extract karke connect karna	Python Pandas, Tableau Data Connector
Data Preparation / Processing	Data cleaning - null hatana, 2W/3W/4W filter, state name format, duplicate remove	Python, Pandas, Data Preprocessing